



Bangladesh Mathematics and Science Olympiad

BdMSO Sample Questions
Science

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Organizer

Society for the Popularization of Science, Bangladesh (SPSB)
Bangladesh Open Source Network (BdOSN)

Building# 758, Green City Centre, Sat Masjid Road, Dhanmondi, Dhaka 1209
info.bdmsso@gmail.com | facebook.com/imsobd | www.bdmsso.org

Multiple Choice Questions

1.

Photosynthesis is a chemical process used by plants, algae, and some bacteria to create their own food that gives them the energy to survive. It is the fundamental process that sustains almost all life on Earth. The reaction takes place inside the chloroplasts of a plant's cells, which are tiny organelles that act like little factories.

What is the primary source of all the energy on Earth?

A. Plants B. Soil C. Sun D. Sea

2.

Light is a form of electromagnetic radiation, and while we can see some types, others are completely invisible to us. Light has some properties that determine its characteristics.

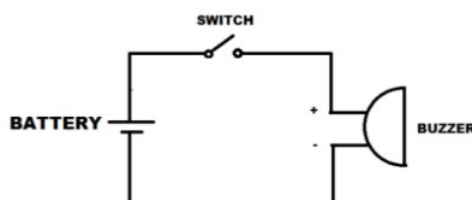
I. Wavelength II. Speed III. Frequency

Which properties of light determine whether we can see it or not?

A. I and II only B. II and III only C. I and III only D. All of the above

3.

The schematic diagram shown below is composed of a battery, a switch, and a buzzer.



In electricity, two concepts are used to describe the flow of current; the conventional current and electron flow. Based on the electron flow notation, which sentence best describes how current flows through a simple circuit when the switch is closed?

- A. Current flows from both the negative and positive sides of the battery to the buzzer.**
B. Current flows from the negative side of the battery to the buzzer to the positive side of the battery.
C. Current flows from the positive side of the battery to the negative side of the battery to the buzzer.
D. Current flows from the positive side of the battery to the buzzer to the negative side of the battery.

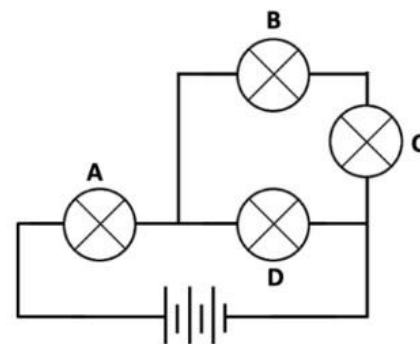
Short Answer Questions

1.
What is the process by which water moves through a plant and evaporates from the aerial parts, mainly through tiny pores called stomata on the leaves?
2.
What is an Eclipse?
3.
List the necessary conditions for heat flow from one object to another.

Brief Answer Questions

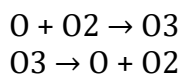
1.
Four identical light bulbs, A, B, C and D are connected as shown on the figure.

A switch S is placed along the circuit such that when S is closed, all the bulbs will light up but when S is opened, only bulbs A and D will light up.

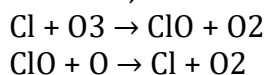


- I. Draw the circuit in your script and locate the possible location of S by drawing a circle on it.
- II. When switch S is closed, identify which light bulb/bulbs will be the- **a.** brightest **b.** least bright

2.
Scientists have discovered a hole in the ozone layer over the Antarctic region. This is largely caused by the release of chlorofluorocarbons (CFCs) in the stratosphere. Ozone (O₃) is constantly produced and depleted in the stratosphere from molecular oxygen (O₂) and atomic oxygen (O).



However, chlorine atom (Cl) from CFCs reacts with the ozone in the following equation:



- I. Explain the importance of the ozone layer to humans.
- II. CFCs are harmful because they introduce chlorine atoms that react with ozone. Even though ozone is constantly produced and depleted, describe and explain how does a single chlorine atom accelerate the depletion of the ozone layer?
- III. Suggest two ways to prevent the further depletion of the ozone layer.